

End of Topic Test – Chapter 5: The Normal Distribution (Test 2) — Mark Scheme

Total: 39 marks

1. Understanding the normal distribution

Loaf mass $\sim N(800, 15^2)$.

- (a) B1 symmetric bell-shaped curve drawn, centred and labelled at mean 800 g. B1 points at one standard deviation each side correctly labelled, 785 g and 815 g. [2]
- (b) B1 recognise 785 and 815 are exactly ± 1 standard deviation from the mean. B1 proportion = 68% (the empirical / 68–95–99.7 rule). [2]
- (c) B2 (or B1+B1) valid explanation, e.g. the mass of a loaf results from many small independent factors (amount of flour and water, proving time, oven temperature variation) combining, which tends to produce a symmetric, bell-shaped distribution clustered around a target mean – a good match to the normal model. [2]

2. Calculating probabilities

$X \sim N(340, 16^2)$.

- (a) M1 $z = \frac{365 - 340}{16} = 1.5625 \approx 1.56$. M1 use normal tables/calculator for $P(Z > 1.56)$. A1 $P(X > 365) \approx 0.0594$ (accept 0.059–0.060). [3]
- (b) M1 both z-values: $z_1 = \frac{310 - 340}{16} = -1.875 \approx -1.88$, $z_2 = \frac{350 - 340}{16} = 0.625 \approx 0.63$. M1 $P(310 < X < 350) = \Phi(0.63) - \Phi(-1.88) = 0.7357 - (1 - 0.9699) = 0.7357 - 0.0301$. A1 ≈ 0.7056 (70.6%) (accept 0.70–0.71). [3]
- (c) B2 (or B1+B1) correct contextual interpretation, e.g. approximately 70.6% (about 7 in 10) of cups dispensed contain between 310 ml and 350 ml of coffee. [2]

3. A job screening test

$X \sim N(58, 11^2)$.

- (a) M1 $z = \frac{45 - 58}{11} = -1.1818 \approx -1.18$. M1 use tables/calculator. A1 $P(X < 45) \approx 0.1190$ (11.9%) (accept 0.118–0.120). [3]
- (b) M1 recognise the top 20% corresponds to the upper tail, using $z = 0.8416$ (given). M1 score = $58 + (0.8416)(11)$. A1 threshold ≈ 67.3 (accept 67.2–67.4). [3]
- (c) B2 (or B1+B1) valid explanation, e.g. test scores are bounded between a minimum (0) and maximum (100 or the maximum available mark), so the true distribution cannot have the unbounded tails of a normal distribution; ceiling or floor effects near the limits mean the real distribution is likely to differ from the normal model there. [2]

4. Interpreting a normal distribution diagram

Fish length $\sim N(28, 4^2)$.

- (a) B1 recognise 24 and 32 are ± 1 standard deviation from the mean. B1 proportion = 68%. [2]
- (b) $M1\ z = \frac{38 - 28}{4} = 2.5$. M1 use tables/calculator for $P(Z > 2.5)$. A1 ≈ 0.0062 (0.62%) (accept 0.006–0.007). [3]
- (c) B2 (or B1+B1) valid reason, e.g. the normal model is simple to apply and gives a good approximation over most of the range (especially near the mean), so it remains useful for practical purposes (e.g. estimating typical catch sizes) even if the model is not perfect for the smallest or largest fish. [2]

5. Interpreting a study using the normal distribution

Reaction time $\sim N(250, 30^2)$.

- (a) B2 (or B1+B1) “normally distributed” means the data forms a symmetric, bell-shaped distribution about the mean (250 ms), with spread described by the standard deviation (30 ms); reaction times close to the mean are most common, and the further a value is from the mean, the less common it is. [2]
- (b) $M1\ z = \frac{305 - 250}{30} = 1.8333 \approx 1.83$. M1 use tables/calculator for $P(Z > 1.83)$. A1 ≈ 0.0336 (3.36%) (accept 0.033–0.034). [3]
- (c) B1 the quotation notes that extreme responses (very fast anticipatory responses and very slow lapses) occur more often than a strict normal model predicts, i.e. the real distribution has heavier tails than the normal curve. B1 this is attributed to factors such as lapses in attention, fatigue and practice effects, which are not part of the idealised model. B1 conclusion: probabilities calculated from the normal model (e.g. part (b)) may underestimate how often very extreme reaction times actually occur. [3]
- (d) B2 (or B1+B1) valid reason, e.g. very fast or very slow reaction times are rarer, so factors such as momentary lapses in attention, fatigue, or anticipation (guessing before the stimulus appears) have a proportionally larger effect on those extreme values, making them less reliable/representative than times near the mean. [2]